



Pre installation

Installation

Operating

**LTL10 Chilled Water System for
GE Twin Magnet MRI Equipment**

**Technical Publication 469
Rev. 2**

The People of Ellis and Watts are committed to set goals and objectives to meet the expectations and needs of our customers. Responsibility and ownership at the point of operation will enhance the success of our mission.

Ellis & Watts International, Inc.
Batavia, Ohio 45103

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REVISION HISTORY PAGE

REVISION NUMBER	DATE	REASON FOR CHANGE	AUTHOR
PRELIMINARY	08/24/01	NEW PRODUCT RELEASE	Jon Richardson
A	10/09/01	REVISED FIXED SITE AND MOBILE SHIP LOOSE, ETL COMMENTS	Jon Richardson
0	2/22/02	REVISED PER GEMS COMMENTS	Jon Richardson
1	2/27/02	REVISED TROUBLESHOOTING TABLE	Jon Richardson
2	3/25/02	REVISED TABLE 6	Jon Richardson

1 INTRODUCTION

1-1 CONTACT ELLIS & WATTS

If a non-service issue arises concerning the equipment or the documentation provided with the chiller you can contact Ellis & Watts in writing at the following address, by phone, or via E-mail at the Ellis & Watts web page.

**Ellis & Watts International
Thermal Sales
4400 Glen Willow Lake Lane
Batavia, OH 45103**

Phone: 513-752-9000

Internet: WWW.ELLISWATTS.COM

GE Field Service will be responsible for service and maintenance issues with this equipment.

1-2 WARRANTY REPLACEMENT PART PROCEDURE

To initiate a warranty claim with Ellis & Watts, contact the Thermal Sales Department. The equipment model number and serial number are required to process a warranty claim. The Thermal Sales representative will issue a Return Goods Authorization (RGA) number. The RGA number must be on the returned part's package.

NOTICE

Items returned without an RGA number will not be processed until the necessary information is obtained from the sender.

1-3 INSTALLATION TOOLS & EQUIPMENT

Item Description	QTY	Used For	Supplied By
Metal Snips	1	Cut banding Straps	Installer
Light oil or soapy water	1 oz	Lubricating Hose Barbs	Installer
LocTite 567 or equivalent	4 oz	Sealing Pipe Threads	Installer
Razor Knife	1	Cutting Hose	Installer
1/2" I.D. Rubber Hose	TBD	Connection Customer Water Supply to Heat Exchanger	Installer

Table 1 Installation Tools and Supplies.

1-3-1 Installation Responsibility Matrix

	Cust. Contr.	GE Service	E&W Service	Who Pays
Once Cooling Cabinets Arrive				
Unload TGWC from Truck	X			Customer
Move TGWC to Equipment Room. (Use seismic Tie down brackets where required by law.)	X			Customer
Connect supplied power cable between TGWC and PDU		X		Customer
Install power and control cable between TGWC and PLC.		X		GE
Once Magnet Arrives				
Install coolant lines from TGWC to 1) Gradient Coil 2) Facility Chilled Water		X		GE
Fill TGWC with Propylene Glycol		X		GE
Start-up TGWC, ensure proper phasing, and no leaks present.		X		GE
Verify unit is controlling to 20°C		X		

1-4 EXPLANATION OF NOTES, CAUTIONS, AND WARNINGS

A **NOTE** is used to provide supplemental information or direction.

CAUTION

A **CAUTION** is used to call particular attention to a step of a procedure, which, if not strictly followed, could result in damage to, or destruction of equipment.

WARNING

A **WARNING** is used to call attention to a particular step of a procedure, which, if not strictly followed, could result in serious injury, or death of personnel.

1-5 SAFETY WARNINGS

WARNING

NO PERSON SHALL REACH INTO THE UNIT ENCLOSURE, WHICH CONTAINS LIVE ELECTRICAL CIRCUITS AS WELL AS ROTATING PARTS, FOR THE PURPOSE OF INSPECTING, SERVICING, ADJUSTING, OR MAINTENANCE EXCEPT IN THE PRESENCE OF SOMEONE CAPABLE OF RENDERING AID. CONTACTING SUCH ITEMS COULD CAUSE SERIOUS BODILY INJURY AND OR DEATH.

WARNING

PERSONNEL WORKING NEAR HIGH VOLTAGES SHOULD BE IN THE PRESENCE OF SOMEONE HAVING RESUSCITATION SKILLS. COMING IN CONTACT WITH HIGH VOLTAGE COULD CAUSE SERIOUS BODILY INJURY AND OR DEATH.

WARNING

TURN OFF AND TAG OUT THE MAIN POWER AT THE DISCONNECT, DURING ANY SERVICING OR MAINTENANCE OPERATIONS. REMOTE STARTING COMPONENTS OR DIRECT CONTACT WITH ELECTRICAL COMPONENTS MAY RESULT IN INJURY TO PERSONNEL.

WARNING

TURN OFF AND TAG OUT THE DISCONNECT TO THE UNIT BEFORE WORKING ON CONTACTORS OR OTHER COMPONENTS IN THE ELECTRIC BOX. WHEN THE MAIN POWER SUPPLY TO THE UNIT IS ON, THE LIVE SIDE OF EACH CONTACTOR CONTAINS HIGH VOLTAGE REGARDLESS OF ANY OTHER SWITCH SETTING. DO NOT ATTEMPT MAINTENANCE PROCEDURES INSIDE EQUIPMENT WITH HIGH VOLTAGE ON. REMOVE POWER AND TAG OUT THE DISCONNECT. GROUND ELECTRICAL COMPONENTS PRIOR TO TOUCHING TO PREVENT BODILY INJURY.

WARNING

STAND CLEAR DURING LIFTING OPERATIONS. OPERATOR ERROR AND/OR EQUIPMENT MALFUNCTION CAN CREATE OPPORTUNITIES FOR PERSONAL INJURY.

WARNING

WHEN AN ABNORMAL CONDITION IS NOTED AND PROCEDURES IN THIS MANUAL DO NOT SPECIFICALLY RELATE TO THE NOTED CONDITION, WORK SHALL BE STOPPED. TECHNICALLY QUALIFIED PERSONNEL SHALL GIVE GUIDANCE ON THE ABNORMAL CONDITION BEFORE WORK CAN BE CONTINUED, TO PREVENT POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE.

2 SHIPPING AND DELIVERY DATA

The General Electric Medical Systems (GEMS) part number 2298100 Water to Water Heat Exchanger Unit (120V-50/60 Hz) is the Ellis & Watts Chiller model LTL10.

2-1 SHIPMENT

For domestic shipment, the TGWC will be shipped to the construction site. International shipment shall be shipped to a collection point where all components for the magnet will be collected and shipped to the construction site.

Unit	Approximate Height	Approximate Width	Approximate Depth	Approximate Weights	
				Domestic	International
TGWC	27 inches (68.58 cm)	38 inches (96.52 cm)	24 inches (60.96 cm)	200 lbs. (90.8 kg)	250 lbs. (113.5 kg)
PLC & Ship loose items	8 inches (20.32 cm)	16 inches (40.64 cm)	14 inches (35.56 cm)	25 lbs. (11.35 kg)	35 lbs. (15.89 kg)

Table 2 Shipping Dimensions

Domestic shipments are packed in a cardboard box and banded on wooden skids. International shipments are domestic packages shipped in crates.

2-2 STORAGE REQUIREMENTS

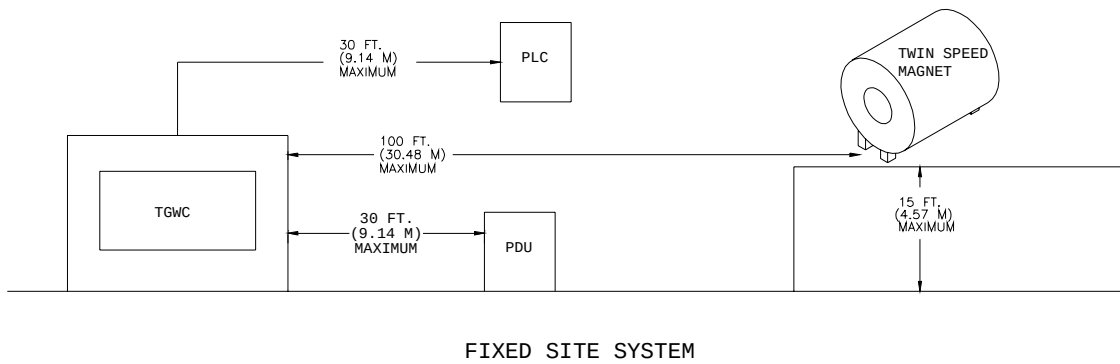
The non-operation environmental conditions are as follows:

Item	Requirement
Ambient Temperature	-25.6°F (-32°C) to 158°F (70°C)
Humidity	5% to 100% RH
Altitude	984 feet (300 m) below sea level 17,060 feet (5,200 m) above sea level
Acceleration	+/- 1.6 G any axis
Magnetic Field	30 Gauss
Note: Do not expose to water (rain, snow)	

Table 3 Environmental Conditions

3 PRE-INSTALLATION REQUIREMENTS

3-1 EQUIPMENT PLACEMENT



The maximum pipe length from the TGWC to the magnet is a total of 100 feet (30.48 m).

Illustration 1 PLACEMENT OF TGWC TO MAGNET

3-2 SITE ENVIRONMENT

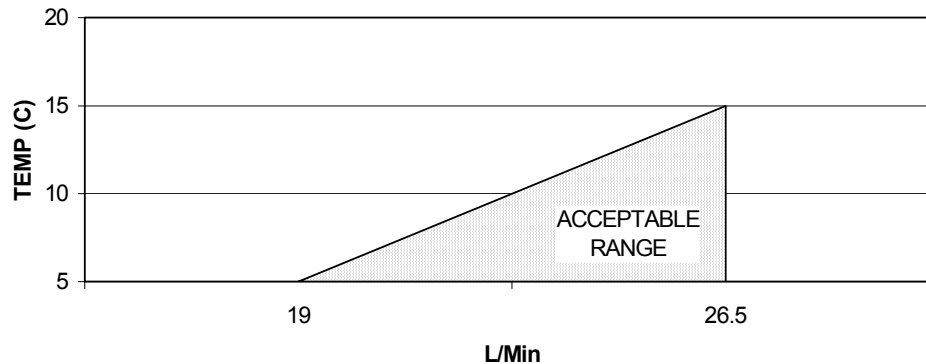
Item	Requirement
Ambient Temperature	41°F (5°C) to 104°F (40°C)
Humidity	10% to 90% RH
Room Cooling Requirements	3,034 Btu/hr
Altitude	984 feet (300 m) below sea level 11,060 feet (3,597 m) above sea level
TGWC Noise Level Maximum	75 dBA in all directions
Magnetic Field	30 Gauss

Table 4 Operating Conditions**3-2-1 Cooling Water**

Item	Requirement
Ph	6.5 – 8.2
Hardness	200 PPM of calcium carbonate
Suspended Matter	Less than 10 mg per liter Less than 150 micron particle size
Anti-freeze	Maximum of 50% glycol in water mixture
Minimum Pressure	137.9 kPa (20 psi)
Maximum Pressure	1033 kPa (150 psi)

Table 5 Water Quality

NOTE: The maximum heat output to the water is 37 Kbtu/hr.

Facility Chilled Water vs. Inlet Temperature Operating Limits**Table 6 Facility Chilled Water Flow vs. Inlet Temperature Operating Limits****3-2-2 Power Requirements**

Item	Requirement
Wire	12 AWG (4 mm ²) 3 conductor /ground
Power Consumption	1.18 KVA
Voltage Requirements	120 VAC $\pm$ 10% 50/60 Hz $\pm$ 3%
Circuit Size	15 AMP
NOTE: Wire is provided in the ship loose kits	

Table 7 Power Requirements**NOTE**

THE LTL10 CHILLED WATER SYSTEM HAS BEEN DESIGNED TO BE CONNECTED TO THE GEMS PDU FOR POWER SUPPLY.

WARNING

THE LTL10 CHILLED WATER SYSTEM SHALL NOT BE MOVED IN ANY DIRECTION ONCE POWER HAS BEEN SUPPLIED TO THE UNIT IN ORDER TO PREVENT PERSONAL INJURY AND/OR DAMAGE TO EQUIPMENT.

3-3 EQUIPMENT CLEARANCE REQUIREMENTS

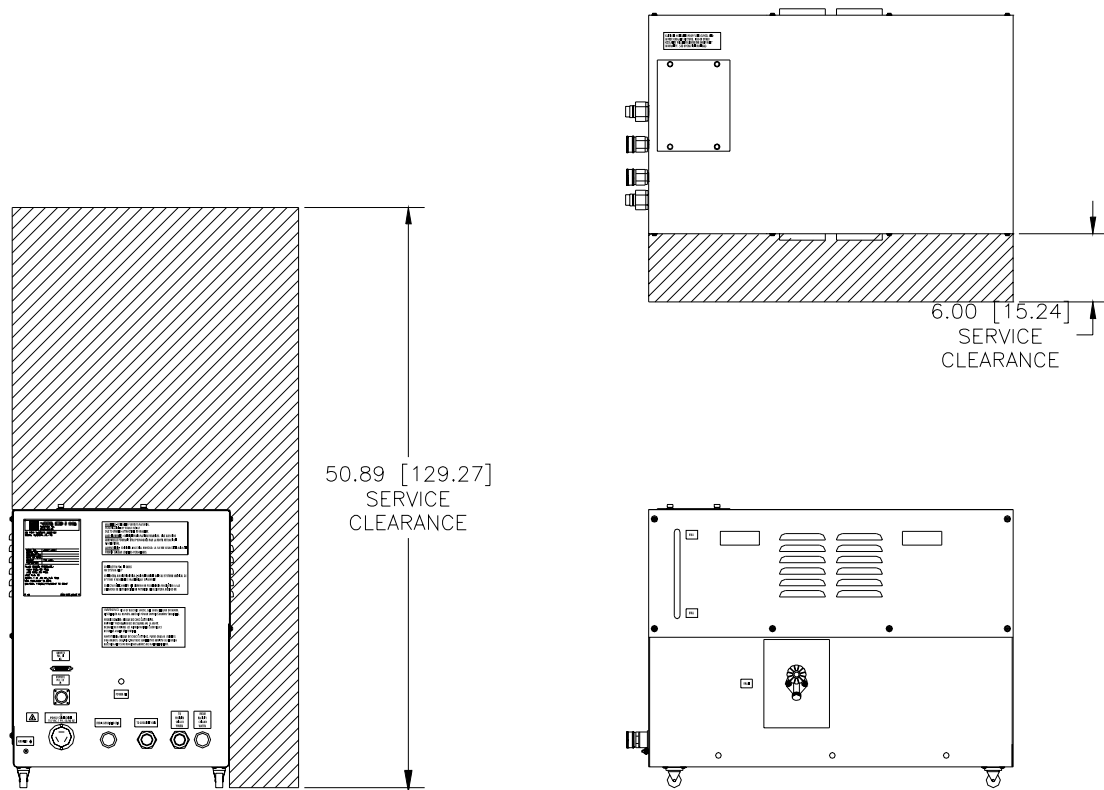


Illustration 2
TGWC CLEARANCE REQUIREMENTS([] = CM)

3-4 COMPONENT DIMENSIONS

Component	Height	Width	Depth	Installed Weight
TGWC	23.67 inches (60.13 cm)	33.94 inches (86.21 cm)	19.66 inches (49.94 cm)	150 pounds
Remote Control Panel	14.44 inches (36.68 cm)	13.46 inches (34.19 cm)	6.15 inches (16.63 cm)	20 pounds

Table 8 Uncrated Dimension

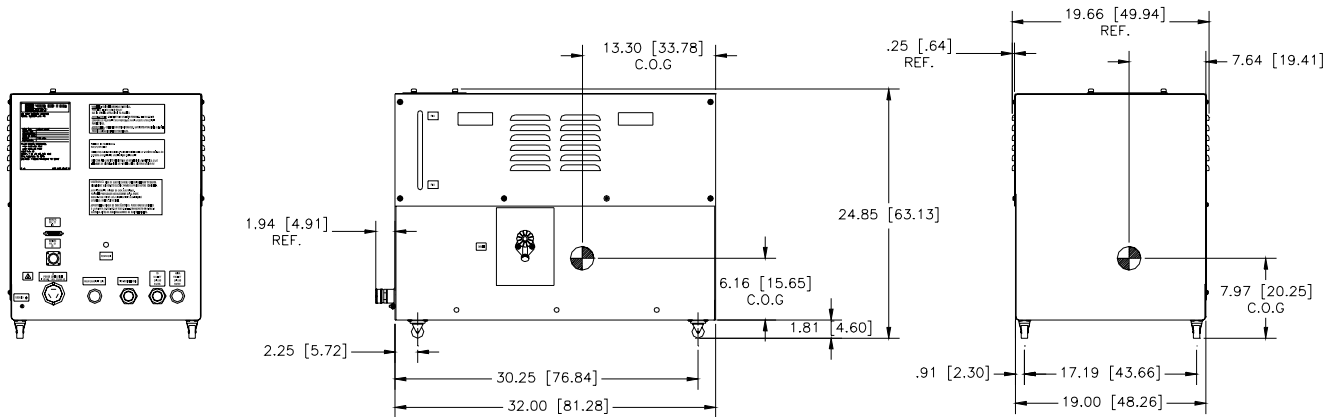


Illustration 3
TGWC DIMENSIONS ([] = CM)

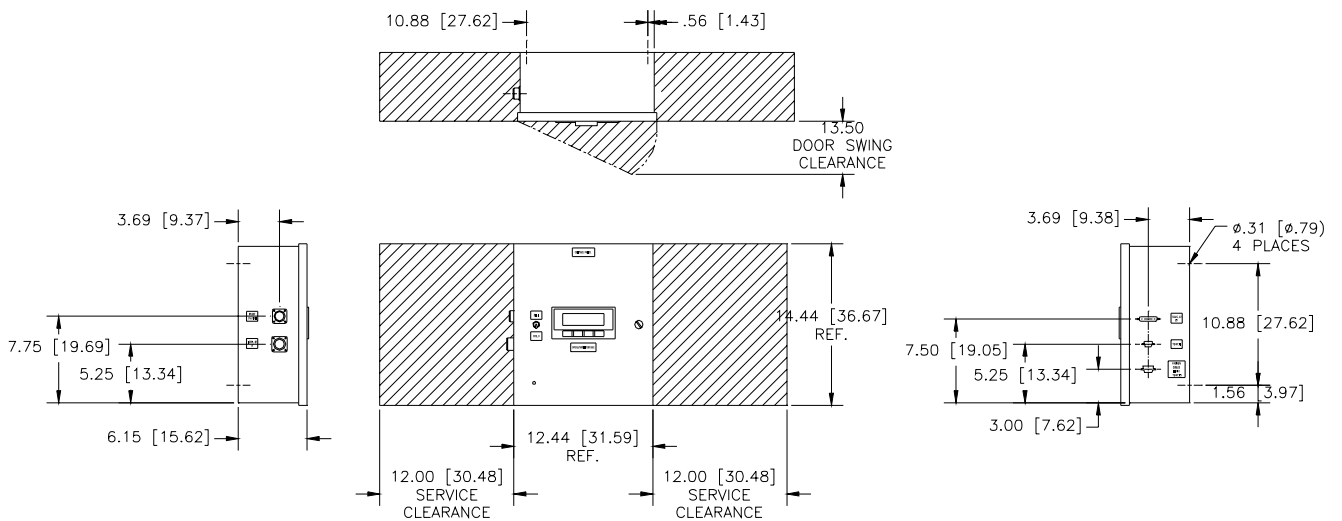


Illustration 4
REMOTE CONTROL PANEL DIMENSIONS ([] = CM)

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3-5 MINIMUM DOOR/HALLWAY SIZES

Component	Hallway/Door Height	Hallway Width *Note	Doorway Width
TGWC	28 inches (71.12cm)	38 inches (127 cm)	23 inches (58.42 cm)
NOTE: Minimum width for 90° turn			

Table 9 Door/Hallway Size**3-6 DOORWAY TREASHOLD CONSIDERATIONS**

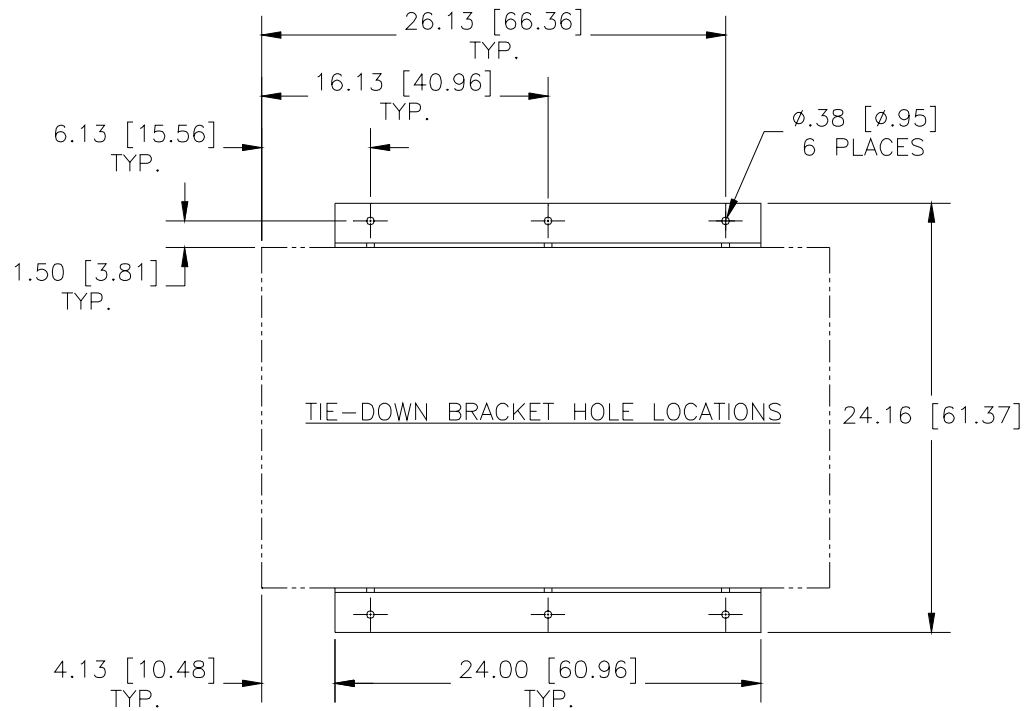
The clearance from the bottom of the TGWC cabinet to the floor is approximately 1.81 inches (4.59 cm) therefore doorway threshold should be as smooth as possible or special considerations have to be made to move cabinet over a threshold height of 1.5 inches (3.81 cm).

3-7 FLOOR LOADING AND WEIGHTS

Equipment	Installed Weight
TGWC	175 lbs. (79.45 Kg)

Table 10 Floor Loading Weights**3-8 RAISED FLOOR CONSIDERATIONS**

The TGWC cabinet is secured to the floor with two tie down brackets. The placement of the tie down brackets need to be taken into consideration when laying out the raised floor framing.



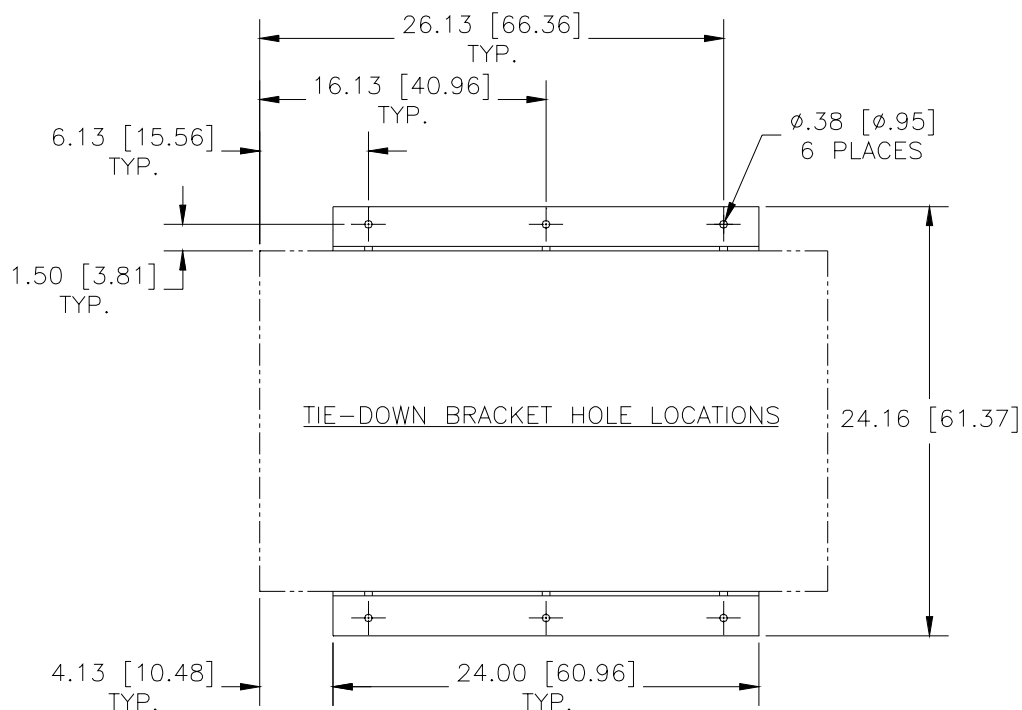
NOTE: USE SEISMIC TIE-DOWN BRACKETS WHERE APPLICABLE.

Illustration 5 SEISMIC BRACKET FOOT PRINT

3-9 ANCHORING AND SEISMIC CONSIDERATIONS

The TGWC cabinet has tie down brackets included in the ship loose parts, which should be used to aid in preventing the cabinet from shifting during a seismic event. Site architects and engineers need to take into consideration the raised floor framing layout in order to have substantial support for the tie down brackets.

Sites requiring seismic anchoring by code should have the site architect and engineer review the response spectra and/or United Builders Code (UBC) for their location and then contact Ellis & Watts International (513-752-9000) for assistance in seismic planning of the site.



NOTE: USE SEISMIC TIE-DOWN BRACKETS WHERE APPLICABLE.

Illustration 6 SEISMIC BRACKET FOOT PRINT

4 UNCRATING PROCEDURES

4-1 UNCRATING AND RECEIVING INSPECTION

NOTE Steps 1 through 3 are for International shipments only.

TGWC

1. Using a #2 Phillips screwdriver, unscrew the wood screws holding on the top of the crate .

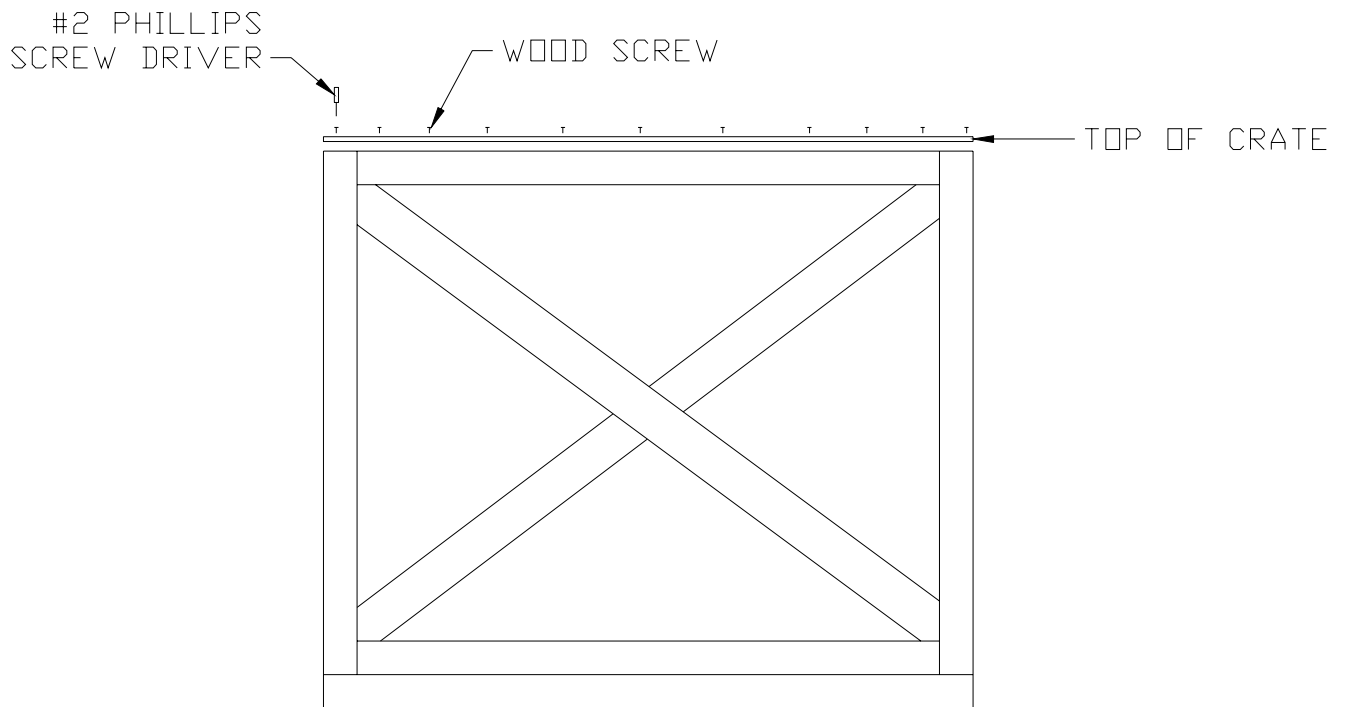


Illustration 7
INTERNATIONAL CRATE EXAMPLE

2. Remove top of the crate and place in an area clear of the uncrating area.
3. Using a prybar carefully remove the front, sides, and back of the crate and place in an area clear of the uncrating area.

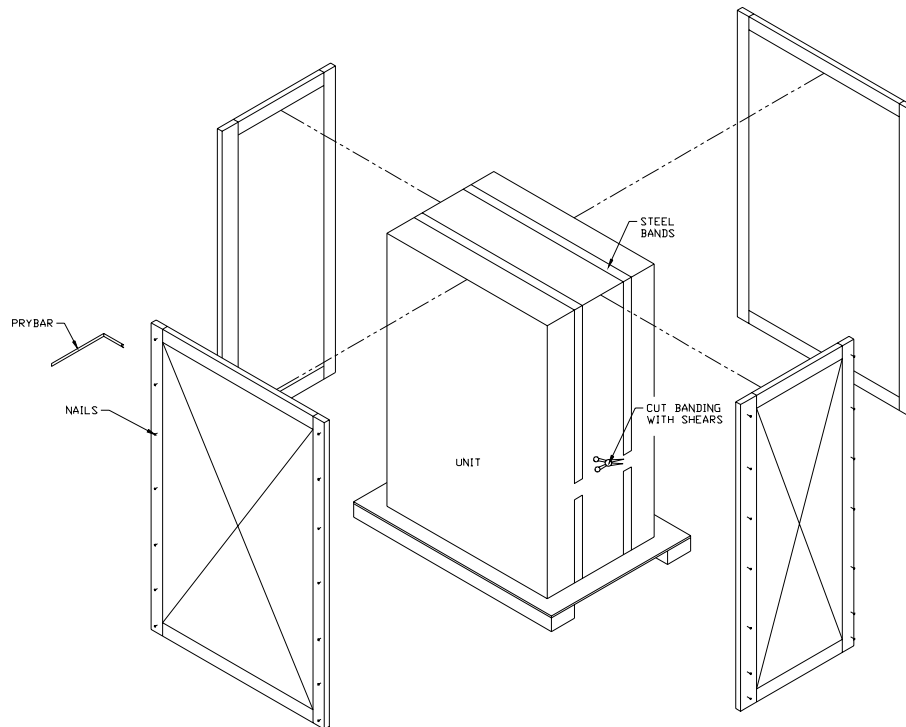


Illustration 8
UNCRATING

CAUTION

4. Banding straps can be under high tension and can whip out when cut.
5. Remove banding by cutting with metal cutting hand shears.
6. Open the cardboard box and remove the plastic wrap.
7. Inspect equipment for any damage. Contact Ellis and Watts and the shipping company as soon as possible and report any discovered damage.
8. Remove the cardboard box containing installation hardware. This box contains the ship loose items and must be stored in a safe location for later use. The box can be inventoried using the list in section 4.2.

4-2 SHIP LOOSE ITEMS

The following tables list the ship loose items that are necessary for the proper installation of the Ellis and Watts equipment. These items are contained in one cardboard box that was shipped with the TGWC. The following tables can be used to verify the inventory of the shipped loose items. Also, the field installation drawings FI-0050, FI-0051, FI-0096, FI-0107, and FI-0109 are included in the ship loose kits.

The TGWC is shipped with the following field installation kits.

No.	Part No.	Ship Loose Items Description	QTY EA
1	J1517-0610	Pump station cable, C5, 50 ft. (15.24 M) Only used on Ellis & Watts vans	1
2	J1517-0613	Power cable C8, 70 ft. (21.34M)	1
3	J1517-0614	Power cable C9, 30 ft (9.144 M)	1
4	FI-0050	Field Installation Electrical/Power/Communication Connections	1

Table 11 Mobile Electrical, Power, And Communication Field Installation Kit

No.	Part No.	Ship Loose Items Description	QTY EA
1	T2636-142	$\frac{3}{4}$ inch Quick disconnect coupler X $\frac{3}{4}$ inch FPT	1
2	J1476-209ZS	$\frac{3}{4}$ inch Hose Barb X $\frac{3}{4}$ inch MPT	2
3	T2590-111	$\frac{1}{2}$ inch Quick disconnect nipple X $\frac{1}{2}$ inch FPT	1
4	62-1-66	Hose Clamp, SST	6
5	J1476-213	$\frac{1}{2}$ inch Hose Barb x $\frac{1}{2}$ inch MPT	4
6	FI-0051	Field Installation piping/hose connections	1
7	T2590-113	$\frac{1}{2}$ inch Quick disconnect coupler X $\frac{1}{2}$ inch FPT	1
8	T2636-140	$\frac{3}{4}$ inch Quick disconnect nipple X $\frac{3}{4}$ inch FPT	1
9	J1520-5050	Propylene Glycol, 50/50 Mixture, 1 Gal GE P/N 2297672	10
10	FI-0107-01	Pump Purge	1
11	FI-0107	Pump Purge Instruction	1

Table 12 Mobile/Fixed Site Piping/Hose Installation Kit

No.	Part No.	Ship Loose Items Description	QTY EA
1	J1517-0201-01	Seismic Tie-Down Bracket	2
2	50-B1T-193	Bolt, Hex head, 3/16 inch -16 X ¾ inch Lg., SST	6
3	50-3E-36	Washer, Flat, 3/8 inch SST	6
4	50-3E-5	Nut, Lock, 3/8 inch, SST	6
5	FI-0096	Drawing, Field Installation, Seismic Tie-Down Bracket	1

Table 13 Fixed Site Seismic Tie-Down Bracket

No.	Part No.	Ship Loose Items Description	QTY EA
1	J1517-0615	Remote Box Power Cable, C14, 30 ft. (9.14 M)	1
2	J1517-0616	Control Cable, C15, 30 ft. (9.14 M)	1
3	J1517-0617	Main Power Cable, 30 ft., (9.14 M)	1
4	FI-0109	Fixed Site Field Installation Electrical/Power/Communication Connections	1

Table 14 Fixed Site Electrical, Power, and Communication Field Installation Kit

5 EQUIPMENT INSTALLATION

5-1 SEISMIC BRACKET INSTALLATION

Verify the parts and quantities from the FI-0096 drawing. Refer to table 13.

Refer to drawing FI-0096 that comes with the ship loose parts, for detailed illustrations.

1. Verify the location selected for the TGWC is in accordance with the following clearance requirements.
 - Minimum of 6 inches (15.24 cm) clearance for the front of the TGWC
 - Minimum of 26.5 inches (67.31 cm) clearance at the top of the TGWC

NOTE: Omit steps 2 - 7 if seismic tie down brackets are not required.

2. Install tie-down bracket to TGWC cabinet with supplied hardware. Refer to Illustration 9.
3. Place 3/8" inch washer onto 3/8" bolt then placing tie-down bracket into position (refer to Illustration 9). Holding bracket in position place bolt through hole in tie-down bracket and TGWC cabinet.

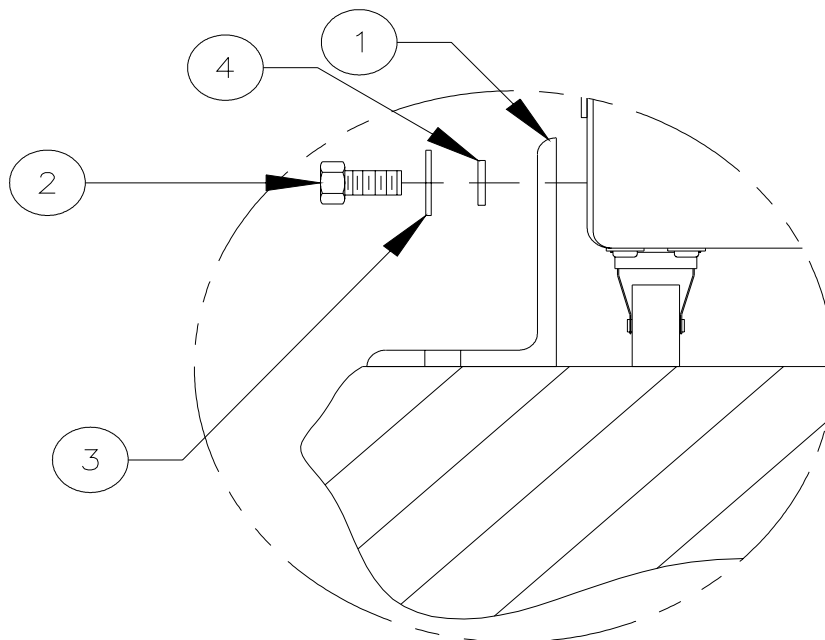
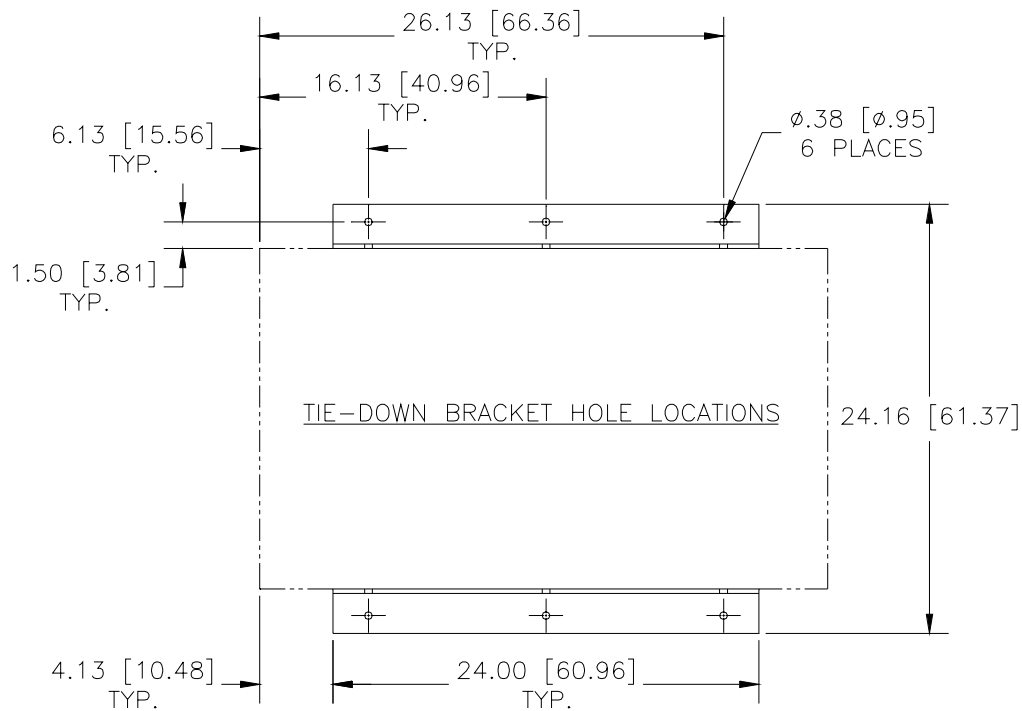


Illustration 9
TGWC TIE-DOWN BRACKET MOUNTING

Refer to Table 13, Items 1 to 4

4. Using a 3/8" wrench, tighten bolt until it is snug against the frame member and then tighten 1/4 to 1/2 turn more.
5. Repeat for the remaining 5 bolts.
6. Match drill floor mounting holes and install floor mounting hardware. (Installer supplied).

The floor mounting hardware is not supplied due to variations in flooring materials. The floor mounting hardware used should be 3/8 inch (M10) hardware that is appropriate for the site selected.



NOTE: USE SEISMIC TIE-DOWN BRACKETS WHERE APPLICABLE.

Illustration 10 SEISMIC TIE-DOWN BRACKET LAYOUT PATTERN

7. Determine torque requirements for the floor mounting hardware and tighten to proper torque rating, taking into account floor panel restrictions.

5-2 ELECTRICAL INSTALLATION

Power Wiring

Circuit and field wiring must be accomplished in accordance to the National Electrical Code, NFPA-70, latest edition and or the Canadian Electrical Code, Part 1, CSA Standard C22.1-1990 and local laws and regulations as they exist.

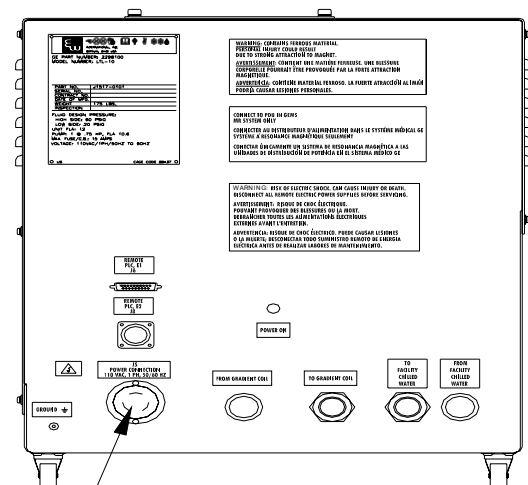
Power Requirement

120 $\pm$ 10% VAC, 50/60 $\pm$ 3 Hz
Full Load current is 1.44 kVA.

Verify the parts and quantities from the FI-0050 drawing for Mobile applications or the FI-0109 drawing for Fixed Site applications. Refer to table 11.

Refer to Mobile drawing FI-0050 or Fixed Site drawing FI-0109, which are ship loose parts, for detailed illustrations.

1. Locate the main power receptacle on the lower left side of the TGWC.



MAIN POWER RECEPTACLE

Illustration 11

MAIN POWER RECEPTACLE LOCATION

2. Install the power cable, insert connector into receptacle and twist to lock in place.

NOTE: Attach the ground wire to the ground stud provided on the TGWC.

5-3 ASSEMBLY OF THE INTERCONNECTING HOSES

Verify the parts and quantities from the FI-0051 drawing. Refer to table 12.

Refer to drawing FI-0051 that comes with the ship loose parts, for detailed illustrations.

1. Determine the length of hoses needed to reach from the TGWC to the Magnet (3/4 inch hose) and the Facility Chilled Water (1/2 inch hose).
2. Cut 2 pieces of 1/2 inch hose to determined length cleanly and squarely with a razor knife.
3. Cut 2 pieces of 3/4 inch hose to determined length cleanly and squarely with a razor knife.
4. Apply a generous amount of Loctite 567 or equivalent to the male pipe threads of each hose barb fitting.
5. Assemble and mark each hose per table.

QUICK DISCONNECT	MARK HOSE AS:
3/4 inch Female	To Gradient Coil
3/4 inch Male	From Gradient Coil
1/2 inch Female	To Facility Chilled Water
1/2 inch Male	From Facility Chilled Water

Table 15 Hose And Quick Disconnect

6. **TIGHTEN EACH THREADED CONNECTION SNUGLY TO PREVENT LEAKS.**

7. Using hose clamps attach each hose assembly to appropriate connection of TGWC.

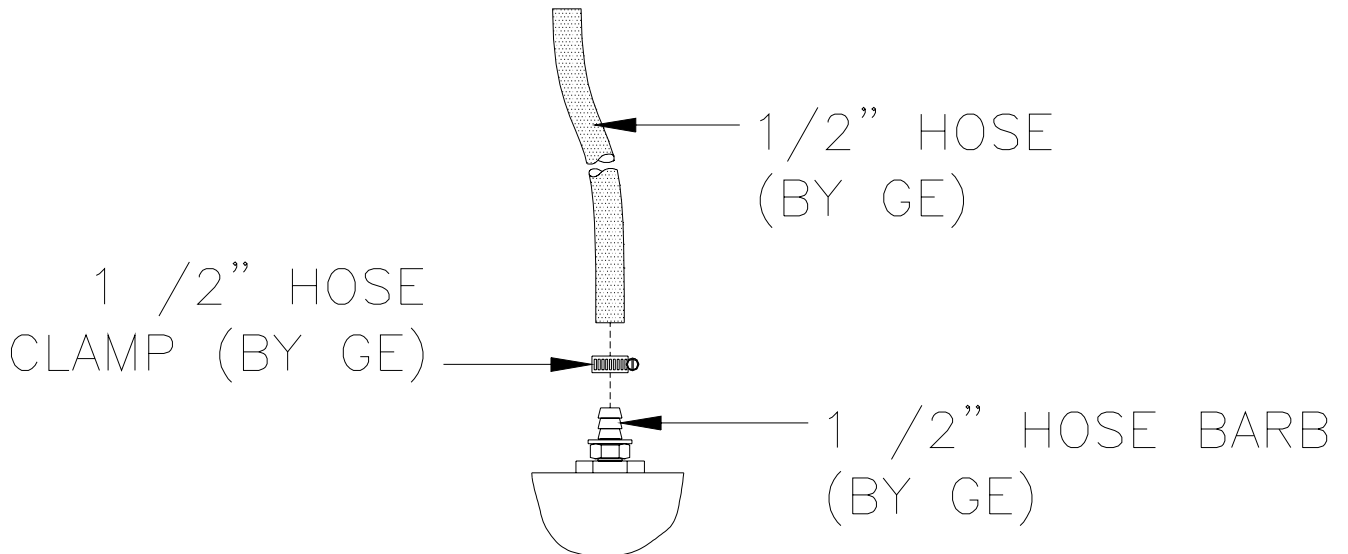


Illustration 12 COOLANT HOSE ASSEMBLIES

8. Route the 1/2 inch hose to the Facility Chilled Water. Route the 3/4 inch hose to the magnet, room.
9. Connect 3/4 inch hose to gradient coil supply and return lines inside the magnet room using hose clamps

6 SYSTEM OPERATION

The following procedures are intended to give instruction on the TGWC system controls and routine operation. Any service must be conducted by GE Field Service Personnel. It is assumed at this point, that the TGWC and associated equipment have gone through initial startup previously and is ready for operation. If the unit has stopped due to a failure, refer to section 6-1-4.

6-1 OPERATING PROCEDURE

6-1-1 Starting Up The TGWC and Associated Equipment

1. Verify that all fluid lines are connected and not leaking. The fluid lines are located on the left side of the unit.
2. Check the coolant level in the reservoir. Verify that the coolant level is between the add and full labels. The reservoir sight glass is located on the front of the unit, (see Illustration 13). If it is not refer to Section 6-1-4.

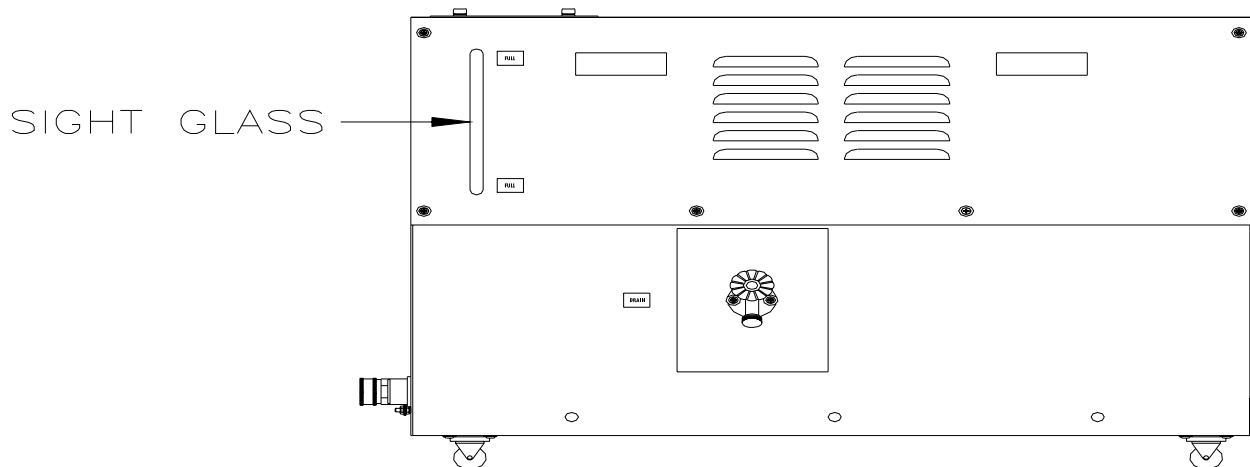


Illustration 13
SIGHT GLASS LOCATION

3. Toggle ON the TGWC circuit breaker, located in the PDU.
4. After 30 minutes of operation verify at the remote control panel that the display is showing proper temperature and flow as shown in table 16.

Equipment	Coolant Flow	Coolant Temperature
Gradient	6.1 GPM +10%	20° +/- 1C

Table 16 TGWC Temperature And Flow Tolerances

6-1-2 Shutting Down The TGWC

1. Verify that the red LED on the Operator Interface on the remote control panel is not illuminated. If the LED is illuminated, check the alarm section of the Operator Interface for any faults and log them before proceeding. Refer to section 6-1-4 for further instructions.
2. Toggle OFF the TGWC circuit breaker located in the PDU.

6-1-3 Maintenance

Conduct routine maintenance every 4 months. This maintenance consists of:

Verifying proper flows and temperatures per table 16 and logging data.

Clean coolant strainer.

Check Remote Control Panel for burned or frayed wiring and loose connections.

Checking all coolant hoses for obvious leaks and for worn areas.

Contact GE for current schedule or approved service provider.

6-1-4 General Troubleshooting

The TGWC chiller system is designed to be monitored by the GE Service Center at all times. Some occasions arise where the operator can assist in troubleshooting until the service provider arrives. In all cases the operator will be **most useful** by checking for and logging any faults on the Operator Interface and noting any abnormal conditions in the equipment room where the TGWC is located. Refer to sections 6-2-1 through 6-2-3 for instruction on how to locate faults on Operator Interface screens.

Problem	Possible Causes	Corrective Action
1.) LTL-10 does not run	a. No power to unit	a. Ensure main power cable, unit power cable, and unit control cable are plugged in securely. See Field Installation Dwg. FI-0109 (Fixed Site) or FI-0050 (Mobile).
	b. ON/OFF Toggle switch set to "OFF" position	b. Turn ON/OFF Toggle switch to "ON" position
	c. Fault condition at PLC	c. Clear alarm at Remote Control Box and allow unit to restart. See Sec. 6-2 of this Manual
2.) No or reduced coolant flow	a. Kink or restriction in coolant lines coming into or going out of the LTL-10	a. Unkink or remove restriction in coolant line. Clear "Low Flow" alarm at PLC.
	b. Obstruction to flow within the coolant lines (i.e. debris)	b. Turn unit off, disconnect coolant lines at the Quick Disconnect fittings, flush LTL-10 unit with deionized water, reconnect fittings, turn unit on.
	c. Leak in coolant line	c. Identify source of leak, turn unit off, tighten leaking fitting or replace leaking hose, turn unit on
	d. Defective Pump Assembly	d. Open Remote Control Box, verify that PLC output 10.02 is illuminated. If so, remove LTL-10 cover, bypass interlock switch, and visually verify if the Pump is running. If still not running, contact GE Service
3.) LTL-10 does not run with power applied	a. Fault condition at PLC	a. Clear PLC Fault at Remote Control Box and allow unit to restart. See Sec. 6.2 of this Manual.
	b. Defective PLC	b. Verify PLC is functional by opening the Remote Control Box door and look for a Green LED light. If not present, replace PLC.
4.) Gradient Coil Supply Temperature not holding to 20°C	a. Chilled water supply to LTL-10 too warm and/or flowrate too low	a. Check temperature and flowrate of chilled water supply, should not exceed 15°C (59°F) at 7 gpm
	b. Two-way valve not controlling properly	b. Check position of two-way valve at Remote Control Box. Two way valve should increase percent open as coolant temperature exceeds 20 degrees C and should decrease percent open when coolant output temperature is less than 20 degrees C. If the valve is not reacting properly, replace unit

TABLE 17 Trouble Shooting Table

6-2 PLC OPERATION

6-2-1 Operator Interface Display Description

The operator interface is located on the Remote Control Panel and displays the model number of the unit, the software revision of the programmed PLC, parameter set points, parameter data, parameter alarm status, and system run time. The interface also allows the user to change the operating set points for the fluid coolant circuit. The upper and lower alarm limits for the various parameters are preset by the manufacturer and are not accessible by service. See Illustration 14 for an example of the layout of the display, user buttons, and indicator lights.

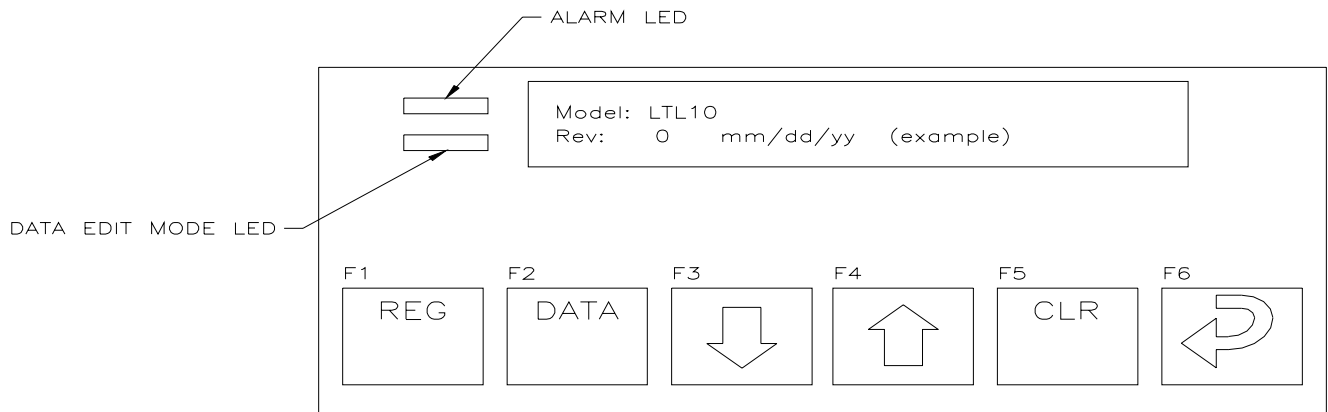


Illustration 14 OPERATOR INTERFACE

6-2-2 Button Description

Button F1 Toggles the display between Menu 1 and Menu 2.

Button F2 Puts the unit into data edit mode for changing the operating set point for supply coolant temperature. The green led will light when the unit is in data edit mode.

Button F3 Menu List Display: Steps through the menu list in the forward direction. Change Set Point: Increases the value shown on the display in 0.1 unit increments.

Button F4 Menu List Display: Steps through the menu list in the reverse direction. Change Set Point: Decreases the value shown on the display in 0.1 unit increments.

Button F5 To extinguish the red alarm status indicator. Note: The operating parameter must be in a normal or no fault mode in order to turn off the red alarm status indicator.

Button F6 Stores new set point changes and exits the data edit mode.

6-2-3 Menu Lists

1. Press the F1 Button to change between Menu 1 and Menu 2.
2. Refer to Table 23, Menu 1 – System Parameter Data, to view the listing of the following types of items: Model Number and Software revision of the unit, operating set points for specific parameters, current parameter data measurement, and total run hours. The first line item on Menu 1 will show the Model Number of the Unit.
3. Refer to Table 24, Menu 2 – System Operating Status, to view the listing of the alarm or failure conditions. The first line item on Menu 2 will show “Supply Fluid Temperature”.

6-2-4 Changing The Set Points

The coolant temperature set points are preset at Ellis and Watts per G.E. specifications. There is a 2°C allowable variance, settable by the operator. If there is a need to change the set point value, then refer to the following steps to perform the change.

1. Press F3 to step forward through the menu until the display shows the coolant temp set point.
2. Press F2 to enter the data edit mode.
3. Press either the F3 or F4 buttons to change the set point value.
4. Press F6 to store the set point values and exit the data edit mode.

Example of Model Number and Revision	
Model Number:	LTL10
Rev.: #	Mo/Dy/Yr
Parameter	Range Value / Unit of Measure (see Note 1)
Supply Temp:	0 to 29.4 Degrees Centigrade
Temp Set Point:	17 to 24 Degrees Centigrade (Factory set at 20°C)
Gradient Flow:	At least 6.1 GPM
2-Way valve % Open:	0 to 100%
Total Run Hours:	0 to 9999 Hrs.
NOTE 1: Unit of measure is set at the factory.	
NOTE 2:	

TABLE 18 Menu 1 – System Parameter Data

Parameter	Display of Operating Mode		
Supply Temp:	Low	Normal	High
Coolant Level:	Low	Normal	
Gradient Flow:	Failure	Low	Normal

TABLE 19 Menu 2 – System Operating Status

TERMS & ACRONYMS

GEMS	General Electric Medical Systems
PDU	Power Distribution Unit
QD	Quick disconnect
RGA	Returned Goods Authorization
EPA	Environmental Protection Agency
MRI	Magnetic Resonance Imaging
UBC	United Builders Code
SST	Stainless Steel
NPTF	National Pipe Thread, Female
PLC	Programmable Logic Control
Liquidtite	Trade name for water tight flexible conduit
LocTite	Brand name for adhesives, sealants, and threadlockers

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MATERIAL SAFETY DATA SHEETS

Dowfrost® Heat Transfer Fluid

Propylene glycol

Dipotassium Phosphate

Deionized Water

Emergency 517 - 636-4400

1. CHEMICAL PRODUCT & COMPANY IDENTIFICATION

Page: 1

24-Hour Emergency Phone Number: 989-636-4400

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

The Dow Chemical Company, Midland, MI 48674

Customer Information Center: 800-258-2436

2. COMPOSITION/INFORMATION ON INGREDIENTS

Propylene glycol	CAS# 000057-55-6	95%
Dipotassium phosphate	CAS# 007758-11-4	<5%
Deionized water	CAS# 007732-18-5	<5%

3. HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW

* Colorless, odorless liquid. Toxic fumes released in fire * situations.

POTENTIAL HEALTH EFFECTS (See Section 11 for toxicological data.)

EYE: May cause slight transient eye irritation. Corneal injury is unlikely.

SKIN: Essentially nonirritating to skin on prolonged contact. A single prolonged skin exposure is not likely to result in the material being absorbed through skin in harmful amounts. Repeated exposures may cause slight flaking, tenderness and softening of skin.

(Continued on page 2)

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low

Product: DOWFROST* HEAT TRANSFER FLUID Product Code: **23545**

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INGESTION: Single dose oral toxicity is low. No hazards anticipated from swallowing small amounts incidental to normal handling operations.

INHALATION: A single prolonged (hours) inhalation exposure is not likely to cause adverse effects. Mists are not likely to be hazardous.

SYSTEMIC (OTHER TARGET ORGAN) EFFECTS: Repeated excessive ingestion may cause central nervous system effects.

CANCER INFORMATION: Did not cause cancer in long-term animal studies.

TERATOLOGY (BIRTH DEFECTS): Birth defects are unlikely. Exposures having no adverse effects on the mother should have no effect on the fetus.

REPRODUCTIVE EFFECTS: In animal studies, has been shown not to interfere with reproduction.

4. FIRST AID

EYE: Flush eyes with plenty of water.

SKIN: Wash off in flowing water or shower.

INGESTION: If swallowed, seek medical attention. Do not induce vomiting unless directed to do so by medical personnel.

INHALATION: Remove to fresh air if effects occur. Consult a physician.

NOTE TO PHYSICIAN: No specific antidote. Supportive care. Treatment based on judgment of the physician in response to reactions of the patient.

5. FIRE FIGHTING MEASURES

FLASH POINT: 215F, 102C METHOD USED: TOC (Tag Open Cup)

(Continued on page 3)

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FLAMMABLE LIMITS LFL: 2.6% @ 100C UFL: 12.5% @ 130C

HAZARDOUS COMBUSTION PRODUCTS: During a fire, smoke may contain the original material in addition to unidentified toxic and/or irritating compounds. Hazardous combustion products may include and are not limited to carbon monoxide, carbon dioxide.

OTHER FLAMMABILITY INFORMATION: Violent steam generation or eruption may occur upon application of direct water stream. Flammable concentration of vapor can accumulate at temperatures above 215.0 deg. F. Liquid mist of this product can burn. Spills of these organic liquids on hot fibrous insulations may lead to lowering of the autoignition temperatures possibly resulting in spontaneous combustion.

EXTINGUISHING MEDIA: Water fog or fine spray, carbon dioxide, dry chemical, foam. Alcohol resistant foams (ATC type) are preferred if available. General purpose synthetic foams (including AFFF) or protein foams may function, but much less effectively. Do not use direct water stream. Will spread fire.

MEDIA TO BE AVOIDED: Do not use direct water stream.

FIRE FIGHTING INSTRUCTIONS: Keep people away. Isolate fire area and deny unnecessary entry. Burning liquids may be moved by flushing with water to protect personnel and minimize property damage. Burning liquids may be extinguished by dilution with water. Do not use direct water stream. May spread fire.

PROTECTIVE EQUIPMENT FOR FIRE FIGHTERS: Wear positive-pressure self-contained breathing apparatus (SCBA) and protective firefighting clothing (includes fire-fighting helmet, coat, pants, boots, and gloves). If protective equipment is not available or not used, fight fire from a protected location or safe distance.

6. ACCIDENTAL RELEASE MEASURES (See Section 15 for Regulatory Information)

(Continued on page 4)

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PROTECT PEOPLE: Clear non-emergency personnel from area.

PROTECT THE ENVIRONMENT: Contain liquid to prevent contamination of soil, surface water or ground water.

CLEANUP: Clean up with absorbent material. Sweep up.

7. HANDLING AND STORAGE

HANDLING: Product on surfaces can cause slippery conditions.

STORAGE: Store below 121 C, 250 F.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

ENGINEERING CONTROLS: Good general ventilation should be sufficient for most conditions.

PERSONAL PROTECTIVE EQUIPMENT

EYE/FACE PROTECTION: Use safety glasses.

SKIN PROTECTION: For brief contact, no precautions other than clean body-covering clothing should be needed. Use impervious gloves when prolonged or frequently repeated contact could occur.

RESPIRATORY PROTECTION: No respiratory protection should be needed.

EXPOSURE GUIDELINE(S): Propylene glycol: AIHA WEEL is 50 ppm total, 10 mg/m3 aerosol only.

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE: Colorless. ODOR: Odorless liquid. VAPOR PRESSURE: 0.22 mmHg @ 20C, 68F
VAPOR DENSITY: 2.62 BOILING POINT: 370F, 188C SOLUBILITY IN WATER: Complete

(Continued on page 5)

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SPECIFIC GRAVITY: 1.050 @ 60/60F, 16C

10. STABILITY AND REACTIVITY

CHEMICAL STABILITY: Thermally stable at typical use temperatures.

CONDITIONS TO AVOID: Avoid temperatures above 121C/250F. Product can decompose at elevated temperatures.

INCOMPATIBILITY WITH OTHER MATERIALS: Avoid contact with oxidizing materials.

HAZARDOUS DECOMPOSITION PRODUCTS: Hazardous decomposition products depend upon temperature, air supply and the presence of other materials.

HAZARDOUS POLYMERIZATION: Will not occur.

11. TOXICOLOGICAL INFORMATION (See Section 3 for Potential Health Effects. For detailed toxicological data, write or call the address or non-emergency number shown in Section 1)

SKIN: The LD50 for skin absorption in rabbits is greater than 10,000 mg/kg.

INGESTION: The oral LD50 for female rats is about 20.3 g/kg.

MUTAGENICITY (EFFECTS ON GENETIC MATERIAL): Results of in vitro (test tube) mutagenicity tests have been negative. Results of mutagenicity tests in animals have been negative.

12. ECOLOGICAL INFORMATION (For detailed Ecological data, write or call the address or non-emergency number shown in Section 1)

ENVIRONMENTAL FATE MOVEMENT AND PARTITIONING: Based largely or completely on data for major component(s). Bioconcentration potential is low (BCF less than 100 or Log Pow less than 3).

DEGRADATION AND PERSISTENCE: Based largely or completely on data for major component(s). Biodegradation under aerobic static laboratory conditions is high (BOD20 or BOD28/ThOD

(Continued on page 6)

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greater than 40%).

ECOTOXICITY: Based largely or completely on data for major component(s). Material is practically non-toxic to aquatic organisms on an acute basis (LC50 greater than 100 mg/L in most sensitive species).

13. DISPOSAL CONSIDERATIONS (See Section 15 for Regulatory Information)

DISPOSAL: DO NOT DUMP INTO ANY SEWERS, ON THE GROUND OR INTO ANY BODY OF WATER. All disposal methods must be in compliance with all Federal, State/Provincial and local laws and regulations. Regulations may vary in different locations. Waste characterizations and compliance with applicable laws are the responsibility solely of the waste generator. THE DOW CHEMICAL COMPANY HAS NO CONTROL OVER THE MANAGEMENT PRACTICES OR MANUFACTURING PROCESSES OF PARTIES HANDLING OR USING THIS MATERIAL. THE INFORMATION PRESENTED HERE PERTAINS ONLY TO THE PRODUCT AS SHIPPED IN ITS INTENDED CONDITION AS DESCRIBED IN MSDS SECTION 2 (Composition/information On Ingredients).

FOR UNUSED & UNCONTAMINATED PRODUCT, the preferred options include sending to a licensed, permitted: recycler, reclaimer, or waste water treatment system.

As a service to its customers, Dow can provide names of information resources to help identify waste management companies and other facilities which recycle, reprocess or manage chemicals or plastics, and that manage used drums. Telephone Dow's Customer Information Center at 800-258-2436 or 517-832-1556 for further details.

14. TRANSPORT INFORMATION

CANADIAN TDG INFORMATION: For TDG regulatory information, if required, consult transportation regulations, product shipping papers, or your Dow representative.

15. REGULATORY INFORMATION (Not meant to be all-inclusive - selected regulations represented)

NOTICE: The information herein is presented in good faith and

(Continued on page 7)

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Product: DOWFROST* HEAT TRANSFER FLUID

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believed to be accurate as of the effective date shown above. However, no warranty, express or implied is given. Regulatory requirements are subject to change and may differ from one location to another; it is the buyer's responsibility to ensure that its activities comply with federal, state or provincial, and local laws. The following specific information is made for the purpose of complying with numerous federal, state or provincial, and local laws and regulations. See other sections for health and safety information.

U.S. REGULATIONS

SARA 313 INFORMATION: To the best of our knowledge, this product contains no chemical subject to SARA Title III Section 313 supplier notification requirements.

SARA HAZARD CATEGORY: This product has been reviewed according to the

EPA "Hazard Categories" promulgated under Sections 311 and 312 of the Superfund Amendment and Reauthorization Act of 1986 (SARA Title 111) and is considered, under applicable definitions, to meet the following categories:

Not to have met any hazard category

TOXIC SUBSTANCES CONTROL ACT (TSCA):

All ingredients are on the TSCA inventory or are not required to be listed on the TSCA inventory.

STATE RIGHT-TO-KNOW: The following product components are cited on certain state lists as mentioned. Non-listed components may be shown in the composition section of the MSDS.

(Continued on page 8)

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REGULATORY INFORMATION (CONTINUED)

CHEMICAL NAME	CAS NUMBER	LIST
1,2-PROPANEDIOL	000057-55-6	PA1

PA1=Pennsylvania Hazardous Substance (present at greater than or equal to 1.0%).

CANADIAN REGULATIONS

WHMIS INFORMATION: The Canadian Workplace Hazardous Materials Information System (WHMIS) Classification for this product is:

This product is not a "Controlled Product" under WHMIS.

16. OTHER INFORMATION

MSDS STATUS: Revised Section 13.

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